



AUGERTORQUE

Manufactured By
Auger Torque Europe Limited



EUROPEAN TRENCHER OPERATORS MANUAL

THE MACHINERY DIRECTIVE**Certificate of Incorporation**

Where Trencher Units are supplied separately, Auger Torque Europe Limited has no control over the manufacture, supply or specification of the mounting frame. To show this and meet with the lawful requirements of the machinery directive no CE mark is applied but a Declaration of Incorporation is issued.

(See page 18 for a copy example)

Declaration of Conformity

Where Trenchers are supplied in conjunction with an Auger Torque Europe Limited manufactured mounting frame to form a Trencher assembly, Auger Torque Europe Limited have control over the suitability of the parts supplied. To show this and meet with the lawful requirements of the Machinery Directive a Declaration of Conformity is issued and a CE mark is applied to the assembly.

(See page I.B.C. for copy example)

DESCRIPTION	PAGE
Forward	Inside Front Cover
Contents	1
Registration	2
Copy Registration form – To be returned for warranty purposes	2a
Identification – Main Parts	3
Introduction	4
Safety Precautions	5
Installing the Trencher	6
Fitting the Trencher – Excavators Double Pin Hitch	7
Fitting the Trencher – Excavators with Quick Hitch	8
Fitting the Trencher– Skid-Steer Loaders	9
Working Procedure	10/11
Transportation	12
Maintenance & Lubrication	13/14
Trouble Shooting – Fault Finding	15/16
Warranty	17
Declaration of Incorporation (Copy Example)	18
Declaration of Conformity (Copy Example)	Inside Back Cover

PRODUCT & MODEL:.....

SERIAL NUMBER:.....

DATE OF MANUFACTURE.....

SUPPLIER / DEALER:.....

DATE SOLD TO SUPPLIER / DEALER.....

DATE SOLD TO ORIGINAL END USER.....



Auger Torque Europe Ltd
Hazelton,
Nr Cheltenham,
GL54 4DX.
England
Tel:+44(0)1451 861652
Fax:+44(0)1451 861660

OWNER OR OPERATOR:

.....

PARENT MACHINE MAKE & MODEL:.....

.....

Note: always quote the serial number in any communication with your supplier/dealer

Copy of Registration Form for Warranty Purposes 2a

PRODUCT & MODEL:.....

SERIAL NUMBER:.....

DATE OF MANUFACTURE.....

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RETURN FOR WARRANTY PURPOSES TO:



Auger Torque Europe Ltd
Hazelton,
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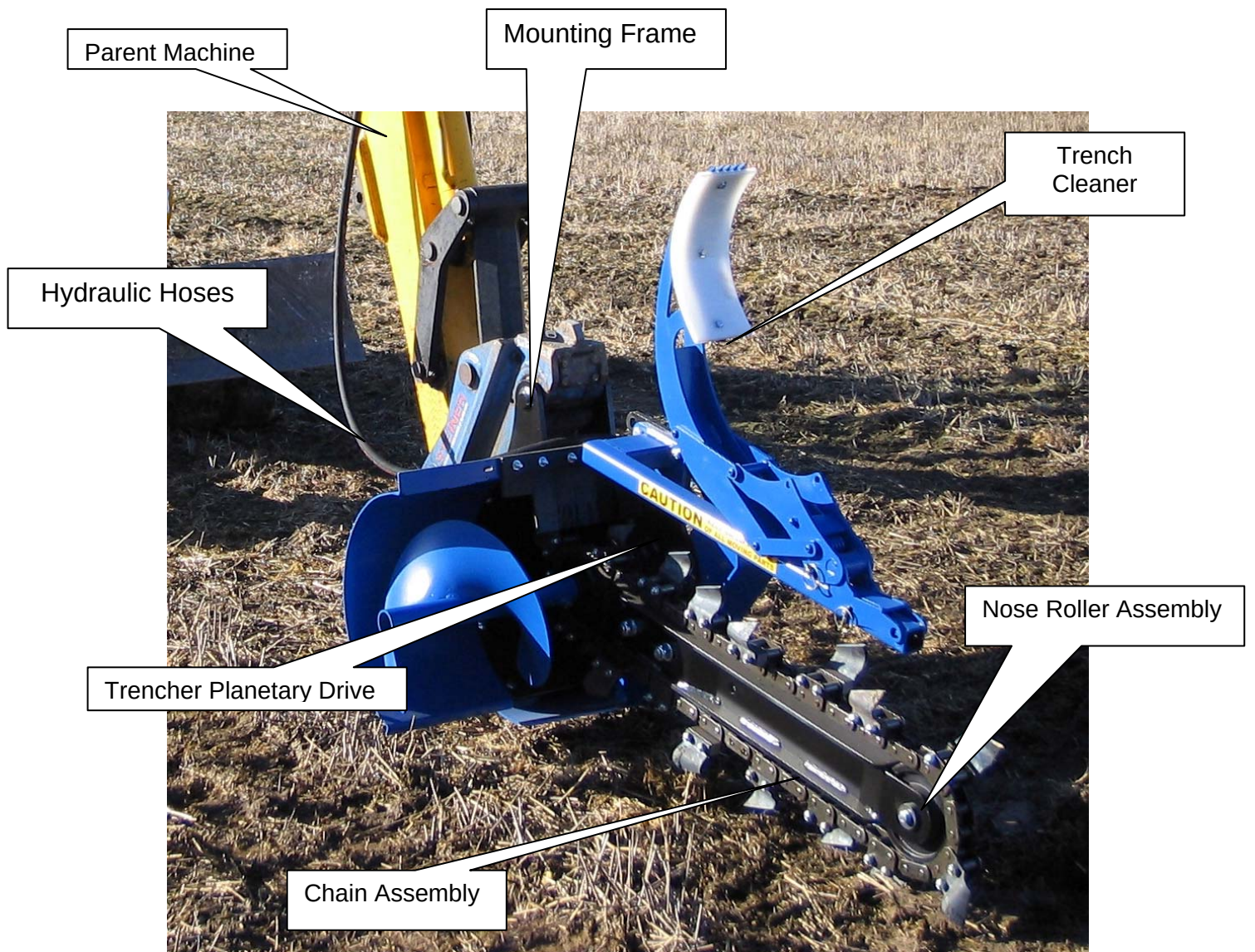
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AUGER TORQUE

Manufactured by:
Auger Torque Europe Limited
Hazleton, Cheltenham
GL54 4DX . ENGLAND
www.augertorque.com

○		Date of manufacture
Model		/...../.....
Serial No.		
Flow (max)	Lpm	○
Pressure (max)	Bar	
Weight	Kg	

SERIAL PLATE
Enter product information for easy reference.



Auger Torque Europe Ltd. and their supplier / dealer thank you for purchasing your new Auger Torque product.

This operating manual has been prepared to enable you to operate the equipment in a safe manner.



Warning, Cautions and Notes

This symbol is used to highlight important messages. When you see this symbol, be alert to the possibility of injury to yourself or others. Carefully read the messages that accompany it.

NOTE:

This operating manual should be used in conjunction with the parent machine's operating instructions

Instruction books should be regarded as part of the machine. They should always be kept safe with the machine for easy and quick reference.

New or extra copies can be obtained from your Auger Torque Dealer or direct from Auger Torque Europe Limited.

Auger Torque Trenchers have been designed for use with specific parent machines along with the Auger Torque range of mounting frames. Provided these are used and maintained correctly, they will provide a safe and reliable method of trenching in the earth.

Auger Torque Europe Limited continually strives to improve and increase its range of products and therefore reserves the right to alter its specifications at any time without notice or obligation. The company accepts no responsibility for discrepancies, which occur between specifications of its machines and descriptions thereof contained in its publications.

When ordering spare parts please quote the serial number of the Trencher, which can be identified by the serial number plate.

SAFETY FIRST



NEVER COMPROMISE ON SAFETY IT COULD CAUSE SERIOUS INJURY OR DEATH. All operators must read and ensure they fully understand all of the safety, operating and maintenance instructions **before** using the Trencher.

If you are in any doubt as to any of the instructions or information provided you must contact your Auger Torque Dealer or Auger Torque Europe Limited **before** attempting to use the Trencher.



Never allow bystanders (including animals) within 6 metres of the work area or allow minors to operate the Trencher Unit.



Never operate or assemble the Trencher without fully understanding the operating instructions of both the Trencher Unit and the parent machine. Auger Torque Europe Limited recommends that you receive dealer instruction before operating the Trencher Unit.

Never operate the Trencher Unit unless you are in good physical condition and mental health.

Never operate the Trencher Unit under the influence of any substance (including drugs & alcohol), which might impair vision, dexterity or judgement.

Always survey the work area before work. Trenching below ground risks electrocution and explosion through contact with unseen hazards such as electricity cables and gas pipes. **See work procedure Page 10**



Always ensure that the parent machine is secure and stable with its engine switched off and the hydraulic circuit is locked out before carrying out any fitting of auxiliary equipment or maintenance work.

Never operate the Trencher Unit with worn, damaged or missing parts. Only use genuine replacement parts.



Never expose fuel or lubricants to any possible source of ignition.



Always protect yourself and the environment. Hydraulic oil, lubricants and exhaust fumes are toxic.



Always tie back long hair and remove jewellery before work. Wear suitable clothing that is close fitting whilst allowing freedom of movement. **Never** wear clothing that could become entangled with the Trencher Unit.



Always protect hands – Trencher parts are sharp. Select gloves that are non-slip to improve grip and ones that protect against contact with oils and greases.



Always protect feet with safety footwear (non-slip soles and steel toe caps are recommended). Trencher Unit parts are heavy and sharp.



Always wear head protection and eye protection when working on the Trencher Unit



Always follow parent machine instruction regarding NOISE protection.



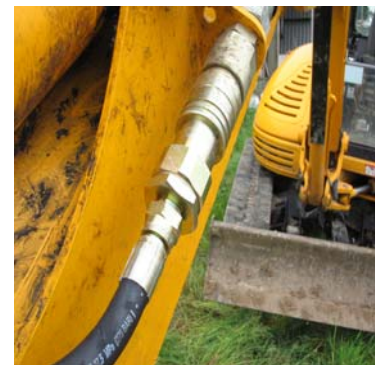
Auger Torque Trencher Units have been designed for use with specific Auger Torque mounting frames and Trencher wear parts. Provided these are used and maintained correctly, they will provide a safe and reliable method of trenching in the earth.

All Auger Torque Trenching Units require a 'flow' and 'return' of hydraulic fluid from the parent machine's auxiliary hydraulic power supply to operate. All Trenchers are reversible but require the host machine to be fitted with a two way flow auxiliary circuit. (Check with parent machine dealer for advice). See fig. 7b for correct chain rotation.



Fig 7b.

Some models of Trencher are supplied with hydraulic hoses, but less hydraulic quick release couplers, which are required for connection to parent machine. These should be sourced locally and be compatible with the auxiliary hydraulic quick release couplers on the parent machine. The parent machine auxiliary hydraulic connections are normally located near the end of the loader arms or excavator dipper.



It is critical that the supply of oil is within the stated limits for the particular Trenching Unit. Flow (l/min) & pressure (Bar).



IF IN DOUBT ASK YOUR AUGER TORQUE DEALER

Always work in pairs (Two skilled operatives) whenever Trencher Unit components are being assembled or disassembled from the parent machine.

Always check parent machine:

- . Is in correct working order
- . Is parked correctly on flat ground
- . The hydraulic circuit is locked out and the engine switched OFF.

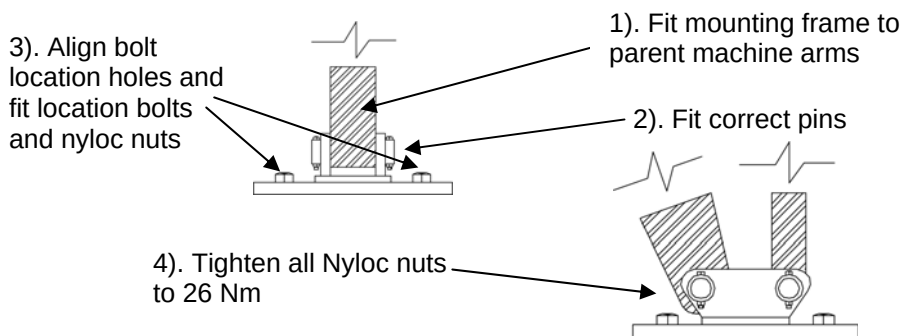
Check that the mounting frame is of the correct model and type to fit the machine.

Ensure mounting frame and attachment points are clean before fitting.

Use suitably rated lifting equipment if required (**see serial plate for weight**)

1

Refer to the parent machines operators' manual for attaching accessories



2

Locate the flow and return hydraulic pipes onto the Trencher Unit



Connect quick release couplers,



Check that the hydraulic pipes are of sufficient length to allow the trencher to articulate during work



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Always check parent machine:

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- . Is parked correctly on flat ground
- . The hydraulic circuit is locked out and the engine switched OFF.

Check that the Trencher is of the correct model and type to fit the machine.

Ensure mounting frame and attachment points are clean before fitting.

Use suitably rated lifting equipment if required (see serial plate for weight)

1

Refer to the parent machine's operator's manual for attaching accessories.



2

Locate the flow and return hydraulic pipes onto the Trencher Unit



Connect quick release couplers.



Check that hydraulic pipes are of sufficient length to allow the Trencher to articulate during work



IF IN DOUBT ASK YOUR AUGER TORQUE DEALER

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Connect quick release couplers,



Check that hydraulic pipes are of sufficient length to allow the Trencher to articulate during work



IF IN DOUBT ASK YOUR AUGER TORQUE DEALER



1



CONSIDER the topography (e.g. risk of subsidence, slope angle, position to embankments and any previous excavation)

NOTE the type of soil and its condition to enable selection of suitable teeth.

ALWAYS carry out a site survey and risk assessment **BEFORE** starting work.

AVOID underground hazards, such as Water/ Gas / Electricity / Communication Lines etc.

If in doubt, detection equipment and professional advice should always be considered before carrying out any work

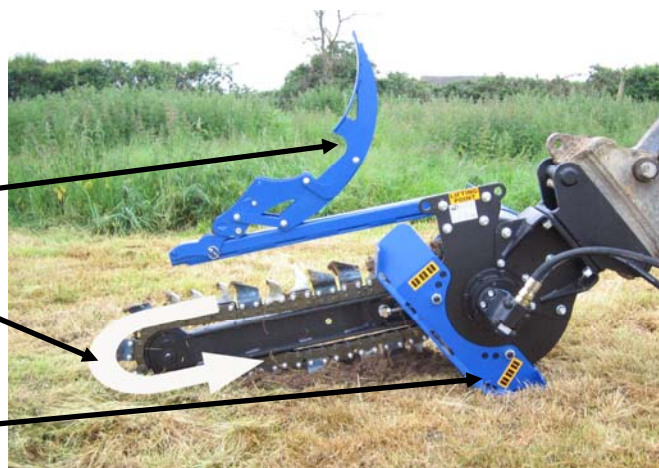
2

ONLY start trenching after checking the site (see section 1 above).

SET Trench Cleaner in its fold back position

ENSURE the chain rotation is correct.

SET Trencher with rear of skid plate firmly on the ground



IF IN DOUBT ASK YOUR AUGER TORQUE DEALER!

3



With Trencher chain rotating **GRADUALLY** extend the parent machine's bucket ram. The trencher will rotate on the skid plate and penetrate the ground



WHEN the bottom of the skid plate is parallel and in firm contact with the ground. **STEADILY** reverse the parent machine, avoid stalling the Trencher Chain.

4

After at least ONE METRE of trenching, raise the trencher and switch off parent machine ensuring the hydraulic system is locked out.

Engage the Trench cleaner system by locking it into its forward position.



IF IN DOUBT ASK YOUR AUGER TORQUE DEALER!

TRANSPORTATION ON PUBLIC HIGHWAYS:

- **ALWAYS** remove the Trencher Unit before driving or transporting the parent machine on public highways.
- **ALWAYS** store the Trencher Unit securely and safely when removed from the parent machine taking special care of the hydraulic hoses and connections.

TRANSPORTATION WITHIN THE JOB SITE:

- **ALWAYS** Disengage the rotation of the Trencher chain, whenever operating the parent machine, whilst the Trencher Unit is not trenching.
- **ALWAYS** operate the parent machine slowly when on site, keeping the Trencher Unit as close to the parent machine and as low as possible.



Recommended Travel Position



IF IN DOUBT ASK YOUR AUGER TORQUE DEALER!



Safety at all times



Ensure environmentally safe disposal of waste oil:

Do not pour down drain!



Avoid Fire or Explosion:

Do not smoke near, or expose lubricants to, any possible sources of ignition

(E.g. fire, electrical sparks or heat sources.)



All lubricants are toxic and potentially carcinogenic (cancer causing).



- **Avoid contact with skin and eyes:**

Wear suitable protective clothing and gloves.

Always use a suitable barrier cream in case of skin contact.



- **Always wear eye protection:**

In the event of skin contact wash with soap and water.

In the event of eye contact wash with water and seek medical advice.

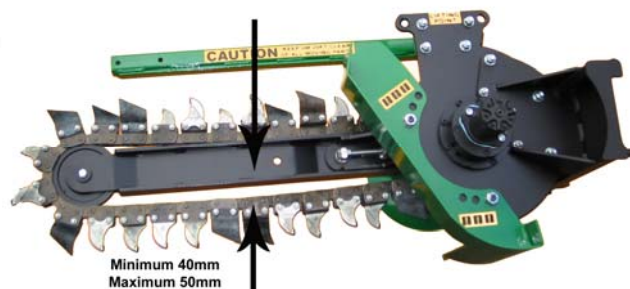


- **Do not ingest:**

If swallowed seek medical advice immediately.

After your first hour of trenching, raise the Trencher unit and check the chain tension.

The chain should be able to move a **MINIMUM** of 40mm **MAXIMUM** of 50mm gap from bottom of boom to chain (as illustrated). Adjust as necessary. Chain tension should be checked regularly.



IF IN DOUBT ASK YOUR AUGER TORQUE DEALER!

Your Auger Torque Trencher Unit features a sealed gear housing filled with gear oil to lubricate the planetary gear set components and bearings within the housing.

Auger Torque Trencher Units are low maintenance, however regular checks for oil leaks and following the service schedules are recommended to ensure a trouble free product.

Daily checks:

Check Chain tension (see page 14).

Check Tooth wear.

Check Condition of nose roller assembly.

Check for oil leaks

Weekly checks:

Check overall condition of the Trencher unit and Mounting frame.

Check drive sprocket for wear.

Check hydraulic hoses for any damage.

After first 500 hours of use or six (6) months:

To maximise life and maintain warranty the Trencher Unit gear oil requires draining and replacing with Castrol SP320 (or equivalent) after the first 500 hours of use or six (6) months from date of purchase – which ever occurs first. (Note this first service is LABOUR FREE provided your Auger Torque dealer carries it out).

Important: To maintain product warranty your Auger Torque dealer must record proof of this first oil change.

Yearly or after every 2000 hours of use (whichever is sooner):

The Trencher Unit gear oil requires draining and replacing with Castrol SP320 (or equivalent) every twelve (12) months or 2000 hours – which ever occurs first. See Parts Manual for volumes.



IF IN DOUBT ASK YOUR AUGER TORQUE DEALER!

Proof of first service

Dealer Stamp

Dealer Name.....

.....

Date Serviced



IF IN DOUBT ASK! - Seek Auger Torque / parent machine dealer for advice & repair.
BE SAFE - only use genuine Auger Torque / parent machine spare parts.

MOUNTING FRAME – ASSEMBLY		
FAULT	POSSIBLE CAUSE	ACTION
Mounting frame does not fit parent machine	Incorrect or non- genuine mounting frame being used	Refer to both this manual <u>and</u> parent machine's operating assembly instructions
	Damaged / worn parts	Repair or replace with genuine mounting frame
MOUNTING FRAME - OPERATION		
FAULT	POSSIBLE CAUSE	ACTION
Excessive movement in locating pins	Incorrect or worn locating pins	Replace with correct new genuine parts
	Parent machine pin location / linkage frame pin location worn	Seek advice from parent machine dealer
	Damaged parts	Seek advice from Auger Torque Europe Ltd / parent machine dealer. Only use genuine spare parts
TRENCHER UNIT - ASSEMBLY		
FAULT	POSSIBLE CAUSE	ACTION
Trencher Unit will not fit mounting frame	Incorrect / incompatible or non-genuine mounting frame / Trencher Unit	Obtain & fit correct and compatible genuine parts
	Damaged parts	Seek advice from Auger Torque dealer. Only use genuine spare parts
Excessive movement in locating pins	Incorrect or worn pins	Replace with correct new genuine parts
FAULT	POSSIBLE CAUSE	ACTION
Trencher drive output shaft does not rotate	No oil flow	Check that parent machine hydraulic two way flow system is operating correctly and has sufficient oil of the correct grade (refer to parent machine operating instructions)
		Check that quick release coupler(s) are correctly engaged to parent machine

AUGER DRIVE UNIT – OPERATION (Cont.)		
FAULT	POSSIBLE CAUSE	ACTION
Trencher Drive output shaft does not rotate	Parent machine pressure relief valve faulty or set too low	Test, reset or replace to parent machine's specification
	Trencher Unit seized	Seek advice from your Auger Torque dealer
	Trencher jammed in ground	Reverse Trencher Chain
Slow Trenching speed / Slow rotation of Trencher Unit output shaft	Insufficient oil flow from parent machine	Check that parent machine hydraulic two way flow system is operating correctly and has sufficient oil of the correct grade
	Incompatible Trencher Unit to parent machine combination	Check specification. Seek advice from Your Auger Torque dealer
	Worn Trencher hydraulic motor possibly due incorrect or dirty oil supply	Seek advice from your Auger Torque dealer. Only use genuine spare parts. Change parent machine hydraulic oil and filter <u>before</u> fitting replacement
Trencher stalls during work	Parent machine pressure relief valve faulty or set too low	Reset / replace pressure release valve to parent machine's specification
	Restricted oil flow	Check for damaged or incorrect hydraulic hoses and connections.
	Blocked hydraulic filter	Change parent machine filter and oil.
	Excessive parent machine pull on Trencher	Reduce Machine Pull on Trencher
	Insufficient parent machine hydraulic pressure	Check that parent machine oil pressure meets with Trencher Unit requirements
	Incompatible Trencher / parent machine combination	Check specification. Seek advice from Auger Torque dealer

All new Auger Torque Europe Ltd. products are warranted to be free from defects in material and workmanship, which may cause failure under normal usage and service when used for the purpose intended.

Auger Torque Europe Ltd. warrants its equipment for a period of eighteen (18) months dating from delivery to original user or a maximum of Twenty One (21) months from date of manufacture which ever occurs first. This warranty covers faulty workmanship and defective parts manufactured by Auger Torque Europe Ltd. Liability is limited to repair or replacement of faulty parts at the discretion of Auger Torque Europe Ltd.

The Warranty does not cover the following: -

1. Normal wear and tear.
2. Faults or failures due to incorrect dealer assembly, pre delivery inspection or installation.
3. Faults or failures due to use under load conditions greater than the design specification.
4. Hydraulic hoses or ground engaging parts such as auger flighting and wear parts.
5. Transportation costs of parts.
6. Consequential loss of any description.

Auger Torque Europe Ltd equipment must be operated in accordance with the recommended procedures and within the ranges specified on the Trencher Unit and stated in the operating manual.

Any claims under this warranty must be made in writing within fourteen (14) days of the fault occurring. Any claims not received by Auger Torque Europe Ltd within 14 days may be deemed invalid.

Auger Torque Europe Ltd is not responsible for nor will accept any charges for work carried out by any unauthorised repairers. Charges, including those for spare parts, will not be accepted unless they have been authorised in writing by Auger Torque Europe Ltd.

Any goods returned to Auger Torque Europe Ltd by the customer, under warranty or for repair, must be freight paid unless authorised in writing by Auger Torque Europe Ltd.



AUGERTORQUE

Manufactured By
Auger Torque Europe Limited



EU Declaration of Incorporation

The responsible person:

Name	
Position	
Company Name	Auger Torque Europe Ltd.
Address	Hazelton, Cheltenham GL54 4DX, England
Telephone	++44 (0) 1451 81465
Fax	++44 (0) 1451 81466

Declares that the product described:

Manufacturer	Auger Torque Europe Ltd.
Model	

Conforms to the Machinery Directive 98/37/EC.

It also complies with the essential health and safety requirements, national standards and the transposed harmonised standards appropriate for this product.

Note: -

This product must not be put into service unless the machinery into which it is to be incorporated has been declared in conformity with the provisions of the Machinery Directive.

Signed by:
(The responsible person)

Dated



AUGERTORQUE

Manufactured By
Auger Torque Europe Limited



EU Declaration of Conformity

The responsible person:

Name	
Position	
Company Name	Auger Torque Europe
Address	Hazelton, Cheltenham GL54 4DX, Eng (0)
Telephone	++44 (0) 1452 36163
Fax	++44 (0) 1452 36166

Declares that the product described:

Manufacturer	Auger Torque Europe Ltd.
Model	
Serial Number	

Conforms to the Machinery Directive 98/37/EC.

It also complies with the essential health and safety requirements, national standards and the transposed harmonised standards appropriate for this product.

Signed by:
(The responsible person)

Dated